

Vaccination for prevention of hearing loss: a scoping review

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Abstract

Plain language summary

Background Infectious diseases in childhood and adolescence are significant and often preventable causes of hearing loss, especially in low- and middle-income countries. We conducted a scoping review to examine the extent, range and nature of available evidence on the role of vaccination for prevention of hearing loss worldwide.

Methods We reviewed the published scientific literature to identify studies providing quantitative information on the relationship between vaccination and hearing loss. We searched four databases: MEDLINE, EMBASE, Cochrane Library and Global Index Medicus. No date, language, or geographical restrictions were imposed. Two independent reviewers assessed eligibility and charted data.

Results Here we show that vaccination may be a key, underexploited strategy for primary prevention of child and adolescent hearing loss. Although the important contributions of rubella and meningitis vaccinations to hearing loss prevention are widely recognised, we identify 26 distinct known or potential infectious causes of hearing loss that are preventable or possibly preventable through vaccination. Notwithstanding, we find a dearth of empirical evidence on the impacts of vaccination on hearing loss prevention. In addition, the review identifies no research from low- and middle-income countries, which bear the overwhelming burden of child and adolescent hearing loss. Finally, it shows that numerous vaccines that address priority infectious diseases relevant to hearing loss are in development and could be brought into use.

Conclusions We recommend strategic investment in research concerning vaccination as a strategy for primary prevention of child and adolescent hearing loss.

Hearing loss is a general term for complete or partial loss of the ability to hear from one or both ears^{1,2}. Usually defined in contrast to normal hearing; that is, hearing threshold of 20

due to hearing loss is concentrated in countries with poor healthcare access, reflecting limited access to assistive hearing devices and necessary care. Attributable to population growth, population aging,

Some infectious diseases (illness from bacteria, viruses or parasites) in childhood and young adulthood can cause hearing loss. Attempting to prevent these infections or lessen their effect could prevent hearing loss. We reviewed the available scientific evidence to learn if vaccines may protect against some of these diseases and thereby help to avoid potential hearing loss. We find that vaccination could be a key strategy not used widely enough, particularly in low-income and middle-income countries. We found evidence suggesting that vaccines protect against a wide range of pathogens (disease causing organisms) related to hearing loss, but very limited observed evidence on the real-world effects of vaccination on hearing loss. We recommend efforts for more research to make better use of vaccines as a strategy for prevention of child and adolescent hearing loss.

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dB or better in both ears, hearing loss may be mild, moderate, severe or profound^{3,4}. Hearing loss currently affects an estimated 1.57 billion people (95% Uncertainty Interval (UI) 1.51 billion to 1.64 billion)—roughly 20% of the world's population—the majority of whom have mild hearing loss⁵. However, ~430.4 million (95% UI 381.7 million to 479.6 million) individuals experience moderate or higher levels of hearing loss likely to negatively impact daily activities and quality of life^{4,5}.

Due to the association between aging and hearing loss, countries with higher median-age populations—largely, higher-income countries—have the highest prevalence of hearing loss⁵. However, the burden of disability

and other factors, a substantial rise in the global prevalence of hearing loss is anticipated, with considerable differentiation across global regions⁵. By 2050, the number of individuals affected by hearing loss is projected to grow by ~56% to 2.45 billion (95% UI 2.35 billion to 2.56 billion); the greatest percentage increases are expected in the African region, followed by the Eastern Mediterranean⁵.

Although hearing loss is commonly related to aging, infectious diseases in childhood and adolescence are significant causes of hearing loss, especially in low- and middle-income countries (LMICs)^{3,5}. Globally, an estimated 70 million children aged 0–15 years are affected by

hearing loss⁵. Although relatively small, this population is especially important, as hearing loss in young children is often more severe and, when unaddressed, may negatively impact spoken language ability, literacy, education, cognitive development and social wellbeing over the lifespan^{5,6}. Over the period 1990 to 2016, hearing loss was the second most prevalent developmental disability among children less than 5 years of age, affecting more than 15 million children annually, overwhelmingly in LMICs⁶. According to the 2021 WHO World Report on Hearing and recent Global Burden of Disease estimates, nearly 60% of childhood hearing loss can be prevented through existing public health measures, such as improved maternal and neonatal care, screening and early management of inflammatory diseases of the middle ear such as otitis media and immunisation against rubella and common causes of meningitis^{4,5}.

Vaccination may have greater scope to contribute to prevention of hearing loss than is currently recognised. Many commonly used vaccines, in addition to those for meningitis and rubella, prevent infectious diseases that can cause hearing loss⁷ and relevant vaccines are in the pipeline. The evidence is dispersed due to the wide range of vaccine-preventable diseases and conditions, and the fact that vaccines are typically studied with respect to primary outcomes such as mortality or serious acute illness. Prevention of hearing loss may be a secondary, understudied outcome of these vaccines.

A preliminary search for existing reviews concerning immunisation and hearing loss, including scoping reviews, systematic reviews and traditional reviews, was conducted in the MEDLINE database (via OVID) on February 17, 2021. Several reviews have been conducted, particularly in paediatric populations, focussing on the causes of hearing loss and its consequences on learning and language acquisition^{8–11}. To date, there has been no comprehensive effort to synthesise knowledge concerning the potential importance of immunisation to reduce hearing loss across the wide range of vaccine-preventable diseases and conditions.

Launched on World Hearing Day 2019, the Lancet Commission on Hearing Loss (LCoHL) convened experts to study how to reduce the burden of hearing loss¹². Because disability due to hearing loss is greatest in countries with lower healthcare accessibility and quality⁵, identification of feasible approaches to strengthen primary hearing loss prevention is especially salient.

To contribute to the Commission's work, we conducted a scoping review to examine the extent, range and nature of available evidence on the role of vaccination in prevention of hearing loss worldwide, to summarise findings and identify gaps in current research, and to develop actionable recommendations for future research. We addressed the following research questions: Which diseases and conditions contribute to hearing loss and are preventable through vaccines currently in use, or vaccines in development? For vaccines currently in use with the potential to reduce hearing loss, has the impact on hearing loss been studied? And for vaccines in the development pipeline? In which geographical areas and in which populations have studies been conducted? What are the most studied target diseases and vaccines, and what are the results to date in relation to hearing loss? Which methodologies and study designs have been used to measure the impact of vaccines on hearing loss? We identify a wide range of vaccines with proven or plausible mechanisms of action to reduce hearing loss, underscoring the importance of immunisation as a strategy for primary prevention of child and adolescent hearing loss. However, we identify only limited observed evidence on the real-world effects of vaccination on hearing loss.

Methods

Protocol and registration

The review protocol follows current guidelines^{13,14} and was prospectively registered with the Open Science Framework on March

30, 2021. The protocol and data extraction forms used to produce this study are publicly available in the Open Science Framework repository at <https://doi.org/10.17605/OSF.IO/EGDSM>¹⁵. Protocol deviations are recorded in the Supplementary Methods. As this study is based on secondary use of publicly available documents containing no personal identifying information, it does not qualify as human subjects research.

Eligibility criteria

Population. All studies conducted on human subjects were eligible for inclusion.

Concept. We hypothesised that specific vaccines could prevent hearing loss in children and adolescents. Studies contributing empirical evidence about the role of vaccination (intervention) for prevention of hearing loss (outcome) were eligible for inclusion.

Interventions. Studies focussing on vaccines with the potential to reduce hearing loss were eligible for inclusion. Amongst these, we distinguished two categories: (1) those in use; (2) those in the development pipeline.

We defined 'vaccines with the potential to reduce hearing loss' using the LCoHL hearing loss aetiology framework, which is a comprehensive list developed by the Commission to identify the causes of hearing loss, broken down by cause category and life stage. We extracted the infectious (bacterial, viral, parasitic) causes of hearing loss from the LCoHL framework and identified vaccines that address these causes. (1) We identified those 'in use' by cross-checking each candidate pathogen in the LCoHL framework against international^{16–18}, and selected regional¹⁹ and national^{20–23} guidelines for routine immunisation, to establish whether a vaccine to counteract the pathogen is, or has been, in use. We considered all vaccines that were ever in routine use, whether or not that use has continued. (2) We identified those 'in the development pipeline' by cross-checking each candidate pathogen in the LCoHL framework against the WHO Vaccine Pipeline Tracker²⁴, which traces vaccine development progress. We focussed on pipeline vaccines in the WHO tracker, as these vaccines are chosen due to their high potential to benefit public health, including in LMICs.

Infectious diseases in the LCoHL hearing loss aetiology framework for which no vaccine was either in routine use or in the WHO pipeline were excluded from the scoping review. However, as vaccines are in continuous development, all diseases and pathogens included in the LCoHL framework but excluded from the scoping review were reconsidered in a horizon scanning exercise (described below). Table 1 summarises inclusion and exclusion decisions.

Outcome(s). The outcome of interest was hearing loss, defined broadly to encompass all types of hearing loss, such as conductive hearing loss, SNHL, and mixed conductive-SNHL, whether unilateral or bilateral³. To be eligible for inclusion, studies required an empirical measure of hearing loss. Suppl 3 characterises the types and severity of hearing loss using definitions adopted in our review⁴.

Context. This scoping review included studies from high-income, middle-income and low-income countries, as defined by the World Bank²⁵.

Types of sources of evidence

Inclusion criteria. Prevention of hearing loss due to vaccination is a little studied topic. Directly searching the scientific literature for this theme was judged non-viable due to the large volume of research on vaccination and the fact that prevention of hearing loss is at best a

secondary outcome of vaccine studies, which made it infeasible to design a sensitive and specific database search strategy. We therefore adopted a two-stage approach to identify published, peer-reviewed primary research studies. In Stage 1, we conducted a structured literature search to identify relevant reviews. In Stage 2, we searched within reviews to identify relevant primary research studies. We applied the same population-concept-context criteria (above) in both stages.

Stage 1: In the first stage, we considered all published, peer-reviewed reviews, including systematic reviews, scoping reviews and traditional reviews²⁶. For vaccines in the WHO development pipeline [modification:

Table 1 | Infectious causes of hearing loss, by vaccine development status and life stage

	Infectious (bacterial, viral, parasitic) pathogens that may	Included in the scoping review cause hearing loss ^a		Excluded from the scoping review Included in the Horizon scan
		Existing vaccine	WHO pipeline in use? vaccine?	
PREGNANCY	Human Immunodeficiency Virus (HIV)		X	
	Lymphocytic choriomeningitis virus (LCMV)			X
	Rubella	X		
	Syphilis			X
	Toxoplasmosis			X
	Zika Virus (ZIKV)		X	
PERI/NEONATAL	Cytomegalovirus (CMV)			X
	Herpes Simplex (HSV)		X	
CHILDHOOD	Cerebral malaria (<i>Plasmodium falciparum</i>) ^b		X	
	Epstein-Barr virus (EBV)			X
	Group B Streptococcus (GBS)		X	
	Haemophilus influenzae	X		
	Lyme disease (<i>Borrelia burgdorferi</i>) ^c	X		X
	Measles, Mumps	X		
	Meningitis	X		
	Non-polio enteroviruses			X
	Pneumococcal disease (<i>Streptococcus pneumoniae</i>)	X	X	
	Respiratory Syncytial virus (RSV) ^b		X	
	Varicella zoster virus (VZV)	X		
ADULTHOOD	Cytomegalovirus (CMV)			X
	Ebola virus (EBOV) ^d	X	X	
	Herpes Simplex 1 (HSV-1)			X
	Herpes Simplex 2 (HSV-2)			X
	Human Immunodeficiency Virus (HIV)		X	
	Lassa virus (LASV)		X	
	Meningitis causing Encephalitis	X		
	Otosyphilis			X
	Ramsay Hunt syndrome/varicella zoster/shingles ^e	X		
	West Nile virus (WNV)			X

^a Extracted from the Lancet Commission on Hearing Loss (LCoHL) aetiology framework.^b Vaccines for malaria and RSV were in the WHO pipeline when this review was initiated. They were approved for use in 2023.^c LYMERix vaccine available in the U.S. was discontinued due to insufficient consumer demand. As it was formerly in use, studies of LYMERix vaccine were eligible for inclusion in the review; however, studies of other vaccines in development against Lyme disease were excluded (included in the horizon scan).^d Licenced vaccine Ervebo not in widespread use although FDA and EMA approved (with exceptions).^e Shingrix vaccine available across the U.S., Canada, and Europe, but not recommended in the WHO vaccination schedule.

and vaccines in use, see Supplementary Methods], we also considered grey literature reports available through the WHO vaccine research and development portal²⁷.

Stage 2: In the second stage, we considered all studies referenced in the reviews and reports identified in the Stage 1; all primary research studies meeting inclusion criteria were selected.

Exclusion criteria. We excluded the following publication types from the first stage of literature searching: reviews of qualitative studies, primary research studies and conference abstracts. Supplementary Table 3 summarises eligibility criteria.

Information sources

Published, peer-reviewed reviews of the scientific literature were identified via four databases: MEDLINE (via OVID), EMBASE (via OVID), Cochrane Library (via Wiley) and Global Index Medicus. No date, language, or geographical restrictions were imposed. Identified records were uploaded into EndNote reference management software (version X9.3.3, Clarivate Analytics, 2017) and duplicates removed. We also considered grey literature reports available through the WHO vaccine product and delivery research portal²⁸. We checked reference lists of included articles and reports to identify additional studies. Authors and subject matter experts were also contacted, as required.

Search

A medical librarian helped to develop the search strategy. Supplementary Table 5 presents a complete search strategy for the MEDLINE (Ovid) database. The most recent search date was 2023-03-19.

Horizon scan

For the infectious causes of hearing loss identified in the LCoHL framework but excluded from this review (i.e. pathogens for which no vaccine is in use, nor in the WHO pipeline, see Table 1) we conducted targeted literature searches in MEDLINE (Ovid) and Google Scholar to ascertain vaccine-development status and evidence related to hearing loss.

Selection of sources of evidence

We conducted screening and selection in two steps using Rayyan software. [Supplementary Methods] In the first step, two independent reviewers (ST, MCPO) screened each title and abstract using a structured eligibility form containing the study inclusion and exclusion criteria. In the second step, for all studies for which eligibility was judged to be possible or uncertain, the same two independent reviewers evaluated eligibility based on the full-text articles. To facilitate consensus, calibration exercises were conducted at each stage. Reviewers resolved disagreements by discussion or appeal to a third reviewer (MJ).

Data charting process

Two reviewers (ST, MCPO) independently conducted data charting using a data extraction template developed for this scoping review¹⁵. Calibration exercises were carried out to adjust the data charting form and improve consistency prior to data extraction. The two primary reviewers came to consensus through discussion or in consultation with a third reviewer (MJ). Data charting was conducted in Microsoft Excel.

Data items

We collected data on article characteristics, and for each individual study extracted, the characteristics of individuals studied (e.g. PROGRESS-Plus characteristics²⁹), population/sample size, type of vaccine(s), hearing outcomes and how they were measured, challenges/ barriers and key findings relating to the scoping review

questions. For all pathogens/ diseases included in the review, we also collated information on vaccine development status (as applicable) and mechanisms of action on hearing loss.

Synthesis of results

We used descriptive measures (counts, frequencies, proportions) to summarise information quantitatively. We also undertook a narrative synthesis aligned with the review objective and research questions.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Results

Infectious causes of hearing loss preventable through vaccination

Based on the LCoHL hearing loss aetiology framework, we identified 26 distinct known or suspected infectious causes of hearing loss that are preventable or possibly preventable through vaccination. [Table 1] Our scoping review included 19 vaccines, comprising 10 vaccines currently in use and 9 vaccines in the WHO development pipeline, addressing 16 infectious causes of hearing loss. A supplementary horizon scan considered 10 infectious causes of hearing loss, for which vaccine development was generally at an early stage.

Search results

Figure 1 describes the process of evidence selection. The search yielded 5 reports and 1183 review articles of which 1085 remained after removing duplicates. A further 1047 articles were removed after title and abstract screening. Of the 38 full-text articles and reports screened for eligibility, 32 were excluded for the following reasons: 23 did not provide evidence on the direct impact of vaccination on hearing loss, 3 articles only considered pathogens excluded from the review scope, 1 article was not a review and 5

articles were not retrieved. Thus, 5 reviews and one report were retained as data sources, from which 9 primary research articles were identified and included in the review. Tables detailing inclusion and exclusion decisions at the full-text stage are provided¹⁵.

Characteristics of included studies

Table 2 describes the nine primary studies included in the review.

The nine studies were published between 1985 and 2013 and presented evidence from six high-income countries (Sweden, Finland, Netherlands, United States, Australia and Japan) located in three World Bank regions (Europe and Central Asia, North America, East Asia and Pacific)^{30–38}. Five articles studied children^{30,31,33,34,36}, one studied adults³⁷ and three articles studying rubella vaccination focussed on infants, school children, and women of childbearing age^{32,35,38}. Six studies used observational methods^{32,34–38}, while three were randomised controlled trials (RCTs)^{30,31,33}.

Vaccines in use with the potential to reduce hearing loss

All nine included studies investigated vaccines in widespread use. We found substantial biological support linking vaccination to prevention of hearing loss for these vaccines (Box 1); however, empirical evidence on hearing loss was available for only three (rubella, mumps and *Streptococcus pneumoniae*) of the seven pathogens.

Three studies^{32,35,38} investigated the impact of rubella vaccination on reduction of hearing loss using observational approaches; all found positive results. These studies shed light on the impacts of Australia's evolving vaccination policy, which began with schoolgirl vaccination in 1971, introduced a universal infant dose of MMR vaccine in 1989,

extended school-based vaccination to both sexes in 1993, and introduced a child MMR booster dose in 1998. In Australia, since 1949, the National Acoustic Laboratories have kept statistics on deafness in childhood, including a congenital rubella deafness (CRD) case registry. Menser et al., reviewed evidence from this registry demonstrating a sharp decline in CRD incidence³². After investigating temporal trends and eliminating competing hypotheses, they concluded that the decline in CRD was due to reduced rubella incidence during pregnancy following introduction of nation-wide rubella immunisation programmes³². Based on case notification data, discharge abstracts and other sources, Cheffins et al. corroborated these findings but noted vaccination coverage gaps in specific sub-populations. The authors found that Australian-born women had higher rubella vaccination coverage than foreign-born women and a corresponding statistically significant difference in rubella seropositivity³⁵. Further analysis of Australia's CRD registry by Gao et al. showed that Australia is approaching rubella elimination³⁸.

Darin et al. studied the aetiology of hearing impairment in children born in Göteborg, Sweden³⁶. Comparing cohorts of children from 1970–1974 and 1980–1984, they found a significant reduction (from 3.8 to 2.0 per 1000 live births ($p < 0.01$)) in the prevalence of all-cause hearing impairment³⁶. Specifically, they observed that cases of SNHL decreased from 3.3 to 1.8 per 1000 live births ($p < 0.01$) and found no cases of congenital rubella or mumps in the latter period. They concluded that the MorbilliParotitis- Rubella vaccination programme introduced in Sweden in 1982 conferred a 12% reduction in hearing impairment among schoolchildren³⁶. Dayan et al. analysed data from a 2006 mumps outbreak in the USA³⁷. The outbreak was caused by gaps in immunity despite high two-dose MMR vaccination coverage³⁷. Of the 4039 mumps patients who met inclusion criteria, 198 (5%) reported complications, including 11 (<1%) with deafness³⁷. Notably, adult patients with clinical mumps who had received at least one dose of the vaccine during childhood presented with transient deafness only, suggesting that childhood immunisation may help to decrease the severity and duration of hearing loss³⁷. Hashimoto et al. conducted a prospective, office-based study to identify the incidence of hearing loss associated with mumps in Japan, where MR (measles - rubella) vaccine (rather than MMR vaccine) is included in the national immunisation programme and mumps is endemic³⁴. Of the 7400 children who underwent hearing ability assessment after clinical onset of mumps, 7 had confirmed hearing loss and none of the 7 had been previously vaccinated against mumps³⁴. The authors found a relatively high incidence of hearing loss in children due to mumps (~1/1000 cases), and concluded that prevention of deafness is an important reason for assuring universal immunisation against mumps³⁴.

Three RCTs^{30,31,33} examined pneumococcal vaccines for the effect of combined pneumococcal conjugate (PCV7) and

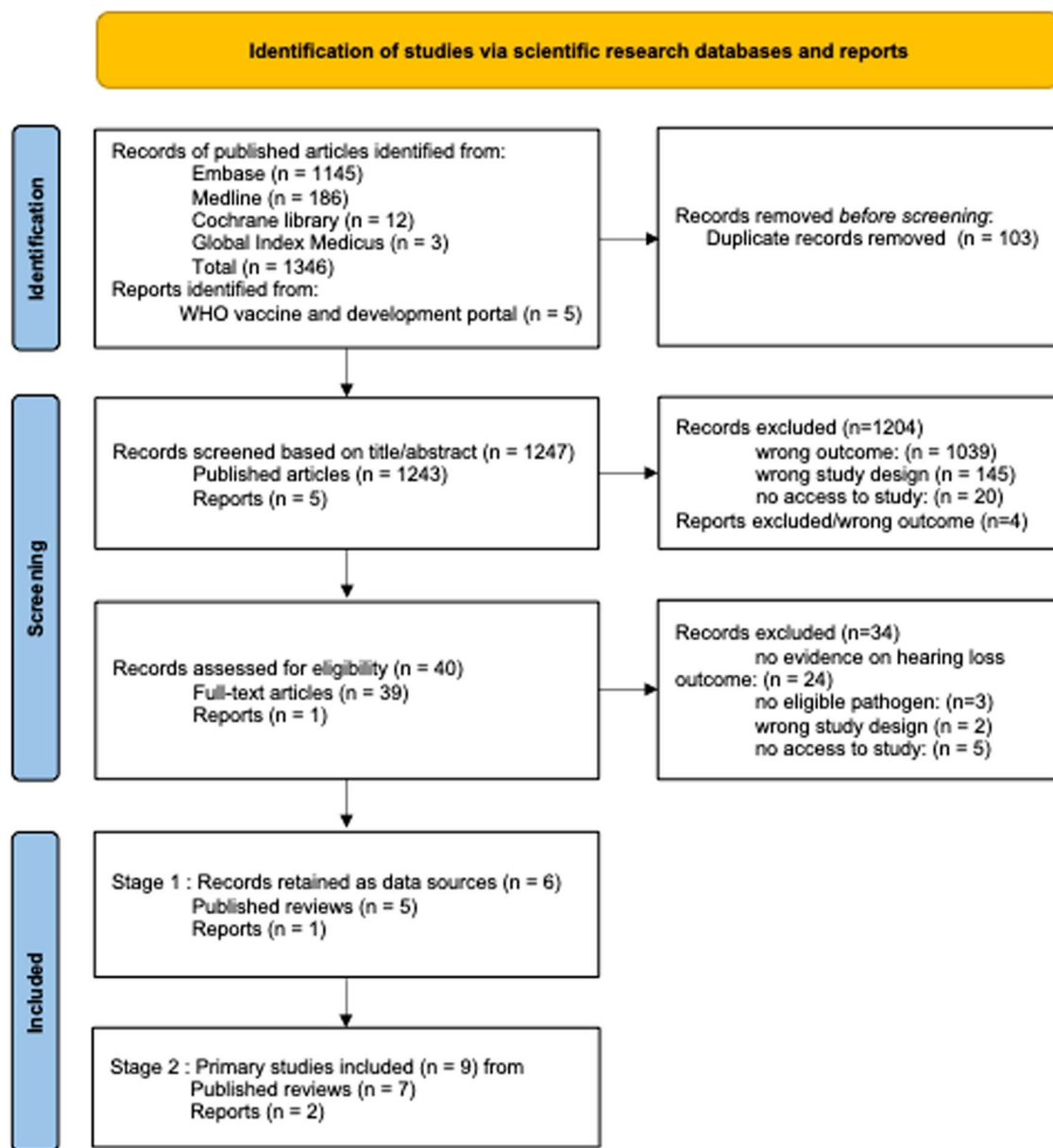


Fig. 1 | Flow diagram of study inclusion.

prevention of recurring otitis media with effusion (OME) in children. All three report negative findings. Straetmans et al. conducted a randomised controlled double-blind trial to evaluate the efficacy of heptavalent pneumococcal conjugate vaccination (PCV7) against acute otitis media in children 2–12 months of age³¹. Although a reduction in OME was observed, the effect was statistically indistinguishable from no effect in overall analyses³¹. Leeta l. studied the efficacy of a regimen of two doses of PCV7 followed by a single dose of the 23-valent pneumococcal polysaccharide vaccine (PPV23) in a randomised double-blind placebo-controlled trial³³. The authors found that combined pneumococcal conjugate and the polysaccharide vaccination had no beneficial effect on OME in children aged 1–7 years with a history of recurrent otitis media³³. Van Heerbeek et al. conducted a randomised controlled trial to assess the

polysaccharide (PPV23) vaccinations on the recurrence of OME among children 2–8 years of age with persistent OME. The authors found no effect of these vaccines in preventing recurrence of OME in this high-risk sub-population.

Measurement approaches varied widely among the 9 included studies. Three studies measured hearing outcomes through different methods of audiometry, either respiratory audiometry in neonatal and early infancy, observation audiometry from the age of 5–6 months, visual reinforcement audiometry from 10 months of age, play audiometry from 2 years 6 months, and auditory brainstem responses (ABR) with threshold determinations^{34,36}. One study also had a test period before hearing loss was suspected, where finger-rubbing audibility tests (rubbing the thumb and index finger beside each ear) or a sound toy for younger children were used³⁴. For the three studies from Australia, cases of congenital deafness were identified through a survey of hearing-aid fittings for rubella-related deafness and not through hearing tests^{32,35,38}. The three RCTs identified OME through

tympanometry (type B or C2) or otoscopic findings (e.g. pneumatic otoscopy), which are not hearing tests^{30,31,33}.

Vaccines in the development pipeline with the potential to reduce hearing loss. For the eight diseases in the WHO vaccine development pipeline (Box 2), we ascertained important biological mechanisms leading to hearing loss; however, we found no empirical evidence related to the impact of the associated vaccines on hearing loss in humans.

Table 2 | Characteristics of included studies

Article	Population	Intervention (vaccine)	Hearing outcomes
Cheffins et al. ³⁵	- Schoolgirls (12–14 years) - Women of childbearing age - Boys & girls 12–15 months	- Rubella immunisation programme starting in late 1970 - MMR vaccine	Declining incidences of CRD according to National Acoustic Laboratories data
Darin et al. ³⁶	Children born from 1980 to 1989, and registered as having hearing impairment by 1991 (90% with SNHL)	Morbili-Parotitis-Rubella vaccination programme in 1982	Comparison of 2 cohorts: 1970–1974 and 1980–1984 found a significant ($p < 0.01$) reduction in hearing impairment: SNHL: 3.3–1.8 cases per 1000 live births
Dayan et al. ³⁷	Adults from national data on mumps cases reported during 2006	Mumps vaccination (one dose, two doses or more)	From mumps cases recorded in 2006, 11 patients were diagnosed with deafness: - unilateral & transient (5) - bilateral & transient (3) - transient (extent unknown) (2) - data missing (1)
Gao et al. ³⁸	- Children 1–12 years - Schoolgirls (10–14 years) - All sexes from 1 to 18 years	Rubella vaccination (with MMR) starting at 1971; booster dose added in 1989	Declining incidences of CRD according to National Acoustic Laboratories data
Hashimoto et al. ³⁴	Children diagnosed with mumps from January 2004 to December 2006: - median age: 5 years	Mumps vaccination: - 8%: 1 dose of mumps vaccine - 92%: no vaccine	Among 24 patients with clinically diagnosed mumps & suspected HL: 7 diagnosed with SNHL (or deafness); 17 with transient HL
Le et al. ³³	Children 1–7 years of age with a history of AOM from the OMAVAX Dutch study conducted between April 1998 and January 2002	Two doses of PCV7 followed by one dose of PPV23	Combined pneumococcal conjugate and polysaccharide vaccination has no beneficial effect on OME in children aged 1 year or older with a history of recurrent otitis media
Menser et al. ³²	- Children with congenital rubella - Females 15–34 years - Schoolgirls (<12 years) - Pregnant women	Rubella vaccination of 10–14-year-old schoolgirls with the Cendehill vaccine strain in 1971	Declining incidences of CRD according to figures provided by the National Acoustic Laboratories
Straetmans et al. ³¹	Children 2–12 months of age from the Finnish Otitis Media Vaccine Trial conducted from December 1995 to March 1999	Pneumococcal conjugate vaccine (7-valent) (PCV7) with the carrier protein nontoxic diphtheria-toxin analogue CRM197 (PncCRM)	PncCRM vaccine did not appear to affect the prevalence of OME
van Heerbeek et al. ³⁰	Children 2–8 years of age diagnosed with OME from April 2000 to March 2002	PCV7 and two forms of (PPV23) (Pneumune and Pneumovax)	The combined PCV7/PPV23 vaccination did not decrease the risk of recurrent bilateral OME

MMR measles-mumps-rubella, CRD congenital rubella deafness (complication of CRS), SNHL sensorineural hearing loss, AOM acute otitis media, OME otitis media with effusion, PCV7 pneumococcal conjugate vaccine (7-valent), PPV23 pneumococcal polysaccharide vaccine (23-valent).

Findings from the horizon scan

The horizon scan found varying degrees of biological support linking pathogens to hearing loss (Supplementary Table 1). Of the seven major syndromes studied, vaccine development was in the pre-clinical stage for three diseases (syphilis, toxoplasmosis, and lymphocytic choriomeningitis virus), and in clinical trials (stages I, II and III) for four diseases (Cytomegalovirus (CMV), Lyme disease, Epstein-Barr Virus, West Nile Virus). Notably, CMV is the most common cause of nonhereditary SNHL in childhood and its link with deafness is well established. Vaccines for CMV are currently in stage II and III trials. (Supplementary Table 1).

Discussion

This scoping review yields four major findings. First, vaccination is an important and under-recognised strategy for primary prevention of child and adolescent hearing loss. Although the important contributions of rubella and meningitis vaccinations to hearing loss prevention are widely acknowledged⁴, our review identified a

multitude of known or suspected infectious causes of hearing loss—26 pathogens—whose deleterious effects on hearing are or could be preventable by vaccination. For a wide range of vaccines in use and at various stages of development, we found plausible biological mechanisms that link specific pathogens to hearing loss and capacity for vaccination to interrupt these pathways. Moreover, there is definitive evidence that vaccines in use for diseases such as rubella, pneumococcal disease and measles prevent clinical outcomes, such as meningitis and acute otitis media, associated with hearing loss

sequelae. Notably, multiple vaccines contribute to prevention of bacterial (pneumococcal disease, *Haemophilus influenzae* serotype b (Hib), *Neisseria meningitidis* and tuberculosis) and viral (measles, mumps) infections that can lead to meningitis. This body of evidence signals that vaccination has much greater scope than previously appreciated as a strategy for prevention of hearing loss due to congenital and infectious causes. Second, notwithstanding, our review found a dearth of empirical evidence on the real-world impacts of vaccines on hearing loss prevention. Relevant studies screened and excluded from our scoping review often reported the pathophysiology of hearing loss and expressed the importance of vaccination to reduce associated complications, but failed to include an empirical measure of hearing loss that would permit assessment of vaccination impact. We found just nine primary studies that met our inclusion criteria, considering only three (rubella, mumps and *Streptococcus pneumoniae*) pathogens, conducted over almost four decades addressing distinct questions in disparate populations using study designs and measurement techniques of varying rigour. Findings related to rubella and mumps vaccination showed a positive population level effect on hearing loss, while results related to pneumococcal vaccination in a high-risk

patient subpopulation were statistically non-significant. Authors' approaches to measuring hearing loss were often not provided or not detailed, rendering comparisons between studies difficult. Third, the review identified no research from LMICs, which bear the overwhelming burden of child and adolescent hearing loss. This is a critical research gap. Fourth, it established that numerous vaccines that address priority infectious diseases and could help to prevent hearing loss are in the development pipeline and could be brought into use. In fact, three critical vaccines to improve child survival and flourishing by combatting malaria³⁹, meningitis³⁹ and RSV⁴⁰ transitioned into use during our study. Our review demonstrates that, as a co-benefit, these also have the potential to prevent hearing loss in children.

These findings are subject to four salient limitations. First, due to the fact that hearing loss is at best a secondary outcome of vaccination, it was judged infeasible to search for studies directly. We hence limited the search to relevant reviews and their references. It is possible that we omitted primary research articles not included in the reviews we assessed. Second, we were unable to locate some candidate articles. Third, the LCoHL hearing loss aetiology framework focusses on

diseases that cause hearing loss. It did not aim to consider diseases for which commonly used therapies may cause hearing loss as a side effect (i.e. ototoxic medicines), which may have led to omission of some relevant vaccines. Fourth, due to the lack of studies meeting inclusion criteria, with the exception of vaccines for prevention of rubella and mumps, we were unable to draw clear pathogen-specific conclusions.

Review findings suggest that vaccination may be a key, underexploited strategy for primary prevention of child and adolescent hearing loss. This topic urgently requires further study. We identified an unprecedented number of vaccines with proven or plausible mechanisms of action to reduce hearing loss, and evidence showing population-level reductions in hearing loss for two (rubella and mumps) of the only three vaccines whose impact on hearing loss has been studied empirically. However, we found a dearth of empirical evidence on the impact of vaccination on hearing loss

prevention overall, and no evidence from LMICs. New, high-quality research on this topic is especially important as the greatest burden of infectious and congenital hearing loss is in LMICs with weaker healthcare access and quality⁵, and childhood deafness is associated

Box 1 | Mechanisms linking vaccination to prevention of hearing loss—vaccines in use^a

Pathogen/disease/	Mechanisms
Rubella ^b	Infection with rubella virus during pregnancy can cause foetal death through miscarriage or stillbirth, or multiple congenital abnormalities through congenital rubella syndrome (CRS). CRS may cause serious birth defects; congenital hearing impairment due to damage to the nerve pathways from the inner ear to the brain (sensorineural hearing loss (SNHL)) is the most frequent. The virus causes direct cochlear damage and cell death in the organ of Corti and stria vascularis. CRS can be prevented by vaccination of women prior to or during pregnancy with a rubella-containing vaccine ^{42,43}
Mumps ^b	Mumps is a viral infection acquired through infected respiratory secretions. Mumps may cause SNHL through mechanisms including atrophy of hair cells in the organ of Corti and stria vascularis, and damage to the myelin sheath around the vestibulocochlear nerve. The involvement of brainstem regions and the 8th nerve is also supposed. Mumps can be prevented by vaccinating children with a mumps-containing vaccine ⁴³ .
Pneumococcus (Pneumococcal meningitis) ^b	Meningitis is an infection of the membranes surrounding the brain and spinal cord. Although various pathogens can cause meningitis, the highest burden is associated with bacterial infection. Most cases are due to four bacteria, including <i>Streptococcus pneumoniae</i> (Sp) or <i>Pneumococcus</i> , which causes pneumococcal meningitis. Hearing loss is most observed in pneumococcal meningitis. Pneumococcal vaccines help to prevent meningitis ^{42,44–48}
Measles	Especially in young children, the measles virus can cause death or severe complications, including hearing loss. Before a vaccine was licensed, deafness due to measles occurred in 4% to 9% of patients. Measles-induced hearing loss is typically bilateral, moderate to profound SNHL. Measles is prevented by vaccinating children with a measles-containing vaccine ⁴³ .
<i>Haemophilus influenzae</i>	<i>H. influenzae</i> is a bacterium which spreads via respiratory secretions and is known to cause bacterial meningitis, with 90% of meningitis cases due to the type B serotype. Similarly to other pathogens causing bacterial meningitis, hearing loss is a well-documented complication ⁴² . Following widespread vaccine administration, ' <i>H. influenzae</i> meningitis now accounts for less than 7% of cases in developed countries but remains an important cause of childhood meningitis in developing countries that do not administer the vaccine' ⁴⁹ .
<i>Neisseria meningitidis</i> /Meningococcus (Meningococcal meningitis)	<i>N. meningitidis</i> is a bacterium which spreads through respiratory secretions. It can cause bacterial meningitis; hearing loss is a well-documented complication ⁴² . Although meningococcal conjugate vaccines are highly effective in preventing infection in children, they were not widely administered in developing countries ⁴⁹ . On October 2, 2023, WHO recommended that all African meningitis belt countries introduce the novel pentavalent meningococcal conjugate vaccine (Men5CV) into routine child immunisation programmes ³⁹ .
Varicella Zoster Virus (VZV)	Varicella Zoster virus is a contagious virus causing a primary infection known commonly as chickenpox which manifests as fever, rash, and pustules. VZV can remain latent and reactivate. When the virus circulates towards the internal auditory canal numerous complications may arise, including SNHL, tinnitus, and vertigo. Vaccination of children against VZV decreases chickenpox incidence by up to 96%. Vaccination of adults 50 years and older with Zostavax can also prevent VZV reactivation ⁴³ .

^a Due to discontinuation of the existing vaccine and new vaccine development efforts, Lyme disease is described in the horizon scan.

^b For these vaccines, empirical evidence on the relationship between vaccination and hearing loss exists and is summarised in the 'results' section.

with a substantial disability burden and unmet needs, especially in these settings⁵. Primary prevention via vaccination is an appealing strategy that is likely to be feasible even in challenging contexts, as routine immunisation systems are in place and actively reinforced, and there are important opportunities to improve coverage of available vaccines and to introduce new vaccines.

New, context-sensitive research will be required to support these aims, especially in LMICs. Vaccination strategies should be informed by local epidemiologic, programmatic and economic considerations, and consider hearing loss prevention among the range of possible co-benefits. We acknowledge that hearing loss is at best a secondary outcome for vaccines, challenging to study due to complex disease aetiologies. We suggest several strategies to improve the evidence base. First, particularly in high-burden settings, we suggest that clinical vaccine trials and other empirical studies consider inclusion of hearing loss outcomes using rigorous designs and standardised measurement approaches. For example, the introduction of new vaccines for meningitis⁴¹ and malaria³⁹ may offer opportunities to

conduct relevant clinical and epidemiological research. Second, economic and modelling studies are often used to inform vaccine introduction and scaling decisions; however, those in LMICs have focussed almost exclusively on benefits related to mortality reduction. We recommend that future studies expand their analytic lens to also consider the impacts of vaccination on morbidity outcomes such as hearing loss, as is done in high-income settings. Focussed research can help to optimise the use of vaccination as a key strategy for primary prevention of child and adolescent hearing loss.

Data availability

The data extraction tables generated and analysed for this study are publicly available in the Open Science Framework repository¹⁵.

Box 2 | Vaccine development stage and mechanisms linking vaccination to prevention of hearing loss—vaccines in the WHO pipeline

Pathogen/disease	Development stage	Mechanisms
Human Immunodeficiency Virus (HIV)	Several HIV vaccine candidates are in Phase I and II clinical trials, including protein subunit, viral vector and ribonucleic acid (RNA) (e.g. mRNA and sRNA) vaccines ⁵⁰ .	HIV causes acquired immunodeficiency syndrome (AIDS). Hearing loss, which may be directly caused by the virus, opportunistic infections in the middle ear and brain, or treatment with ototoxic medications, figures among the possible sequelae ⁴³ .
Zika Virus (ZIKV)	Several ZIKV vaccines are in Phase I and II trials, including DNA and purified inactivated, mRNA, viral vector and live attenuated vaccines ^{51–53} .	Zika is a vector-borne disease primarily acquired through Aedes mosquitoes, known to affect the auditory system. Auditory impairment of peripheral origin is suspected, through cochlear or neural involvement ^{54,55} .
Plasmodium falciparum (Malaria)	At review outset, there were no licensed malaria vaccines. Two vaccines - RTS/S subunit ^{56,57} and R21/Matrix-M ³⁹ - are now approved by WHO for malaria prevention in children.	Malaria is a vector-borne disease spread by Anopheles mosquitoes carrying the parasite Plasmodium falciparum to human hosts. Especially in children, a complication known as cerebral malaria can lead to neurocognitive impairment with ensuing hearing loss. Suspected mechanisms include loss of function from the auditory cortex to the cochlear hair and microcirculation failure ^{58–60} .
Respiratory Syncytial Virus (RSV)	At review outset, there were no licensed RSV vaccines ⁶¹ . There are now 3 FDA-approved RSV vaccines, two for individuals over age 60 ⁶² and one given to pregnant women to protect infants younger than 6 months ⁴⁰ .	RSV is a common respiratory virus often causing viral otitis media, an inflammation or infection in the middle ear. Chronic otitis media can lead to conductive and SNHL. As RSV is the virus most frequently recovered in the middle ear, an effective vaccine may reduce acute otitis media incidence ^{63–65} .
Group B Streptococcus (Streptococcus agalactiae)	Two types of vaccines for GBS are being developed through preclinical and Phase I/II trials: CPS conjugate vaccines and protein-based vaccines ⁶⁶ .	GBS is a leading cause of neonatal sepsis and meningitis. Meningitis can cause SNHL through damage to the cochlear hair cells or the auditory nerve, through bacteria, toxins, or immune response ^{42,67} .
Lassa Virus (LASV)	All vaccine candidates are in preclinical trials ⁶⁸ .	A potentially fatal viral haemorrhagic fever, LASV may also cause neurological sequelae, with approximately one third of survivors developing SNHL. Hearing loss is likely due to an immune-mediated reaction rather than direct viral damage ^{69–71} .
Ebola Virus (EBOV)	There are 4 licensed vaccines ⁷² , including one prequalified by WHO for use in high-risk countries ⁷³ . Several candidates are in phase III clinical trials ⁷² .	An often fatal haemorrhagic fever, EBOV has multiple sequelae including hearing loss. Mechanisms of pathogenesis remain unknown, but animal studies suggest EBOV penetration into cerebrospinal fluid through the cochlear aqueduct ⁷⁴ . Hearing loss may involve virus-induced tissue damage ^{74–76} .
Herpes Simplex Virus (HSV)	Numerous vaccine candidates are in preclinical and Phase I trials, with Phase IIa planned ^{43,77–83} .	HSV commonly causes recurrent oral and genital ulcerations. Mechanisms of pathogenesis for hearing loss include invasion of the cochlear nerve, or triggering an immune response.

Abbreviations

ABR	auditory brainstem responses
AOM	acute otitis media
CMV	cytomegalovirus
CRD	congenital rubella deafness (complication of CRS)
CRS	congenital rubella syndrome
EBOV	Ebola Virus
FDA	US Food and Drug Administration
GBS	Group B Streptococcus
HSV	Herpes Simplex Virus
HIV	human immunodeficiency virus
LASV	Lassa fever
LCoHL	Lancet Commission on Hearing Loss
LMICs	low- and middle-income countries
MMR	measles-mumps-rubella
MR	measles-rubella
OME	otitis media with effusion
PCV7	pneumococcal conjugate vaccine (7-valent)
PPV23	pneumococcal polysaccharide vaccine (23-valent)
RCT	randomised controlled trial
Ribonucleic acid	RNA
RSV	Respiratory Syncytial Virus
SNHL	sensorineural hearing loss
UI	Uncertainty Interval
VSV	vesicular stomatitis virus
VZV	Varicella Zoster virus
ZIKV	zika virus

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Author contributions

M.J., E.B. and B.W. conceived the study. S.T. and M.C.P.O. did the literature search, study selection and extracted the relevant information, in consultation with M.J. M.J. accessed and verified the data. S.T., M.C.P.O. and M.J. synthesised the data. M.J. wrote the original paper draft, with contributions from S.T. M.J. supervised the overall work. All authors critically revised successive manuscript drafts. All authors had full access to all the data in the study, read and approved the final manuscript, and had final responsibility for the decision to submit for publication. All authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Competing interests

The authors declare no competing interests.

Additional information

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